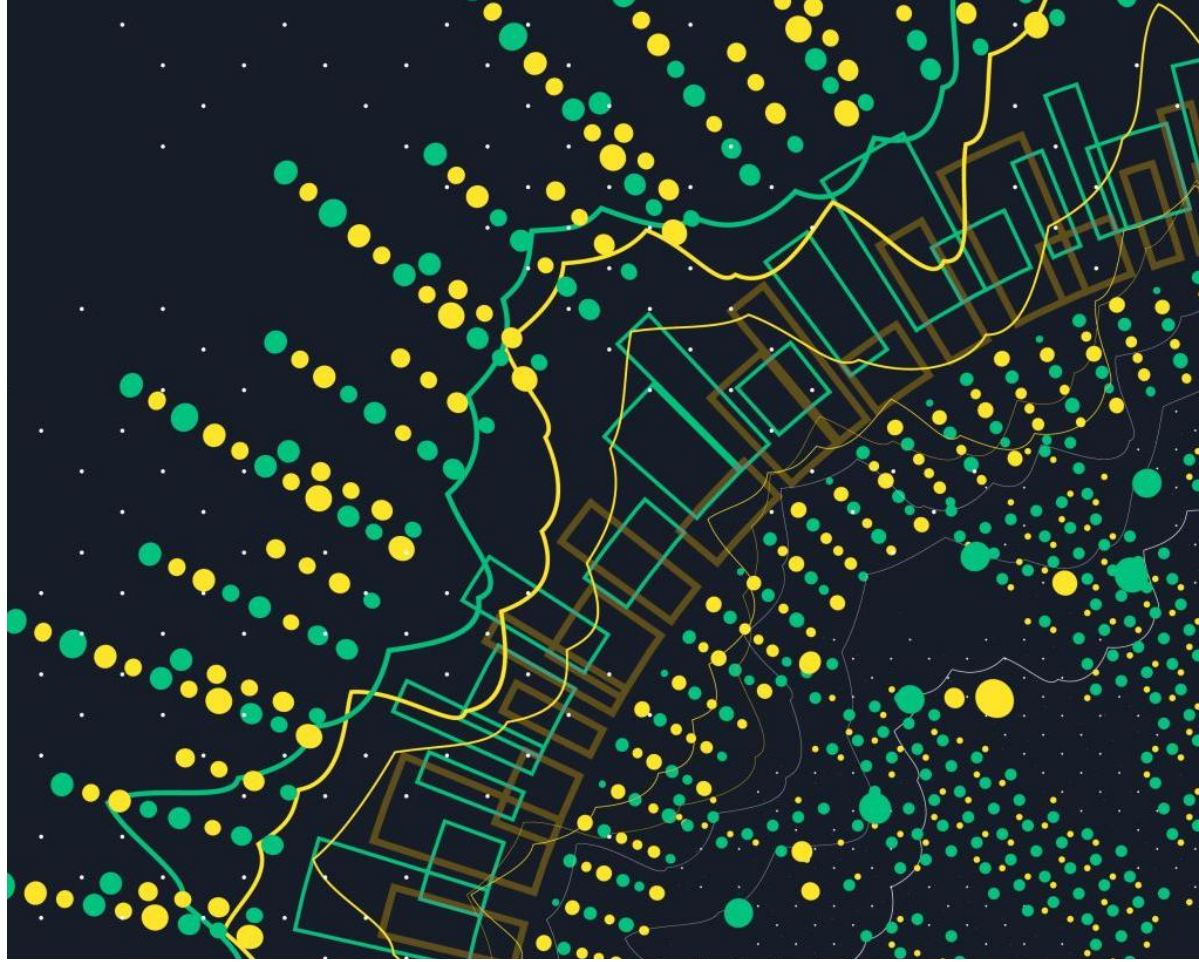
MOBILE DEEP LEARNING INFERENCE FRAMEWORK FOR ANDROID-BASED SHORT-CIRCUIT TRACE CLASSIFICATION



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INTRODUCTION

01

Electrical fires often caused by short circuits

02

- Need to distinguish PSCT vs SSCT

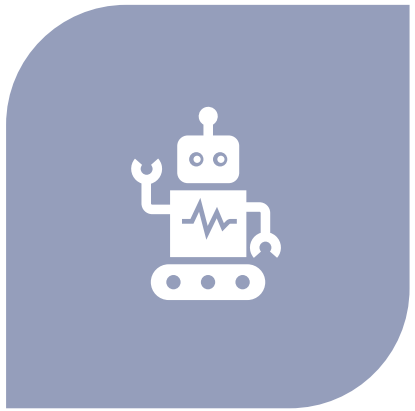
03

- Traditional methods are manual and subjective

04

- AI enables automated classification

OBJECTIVE

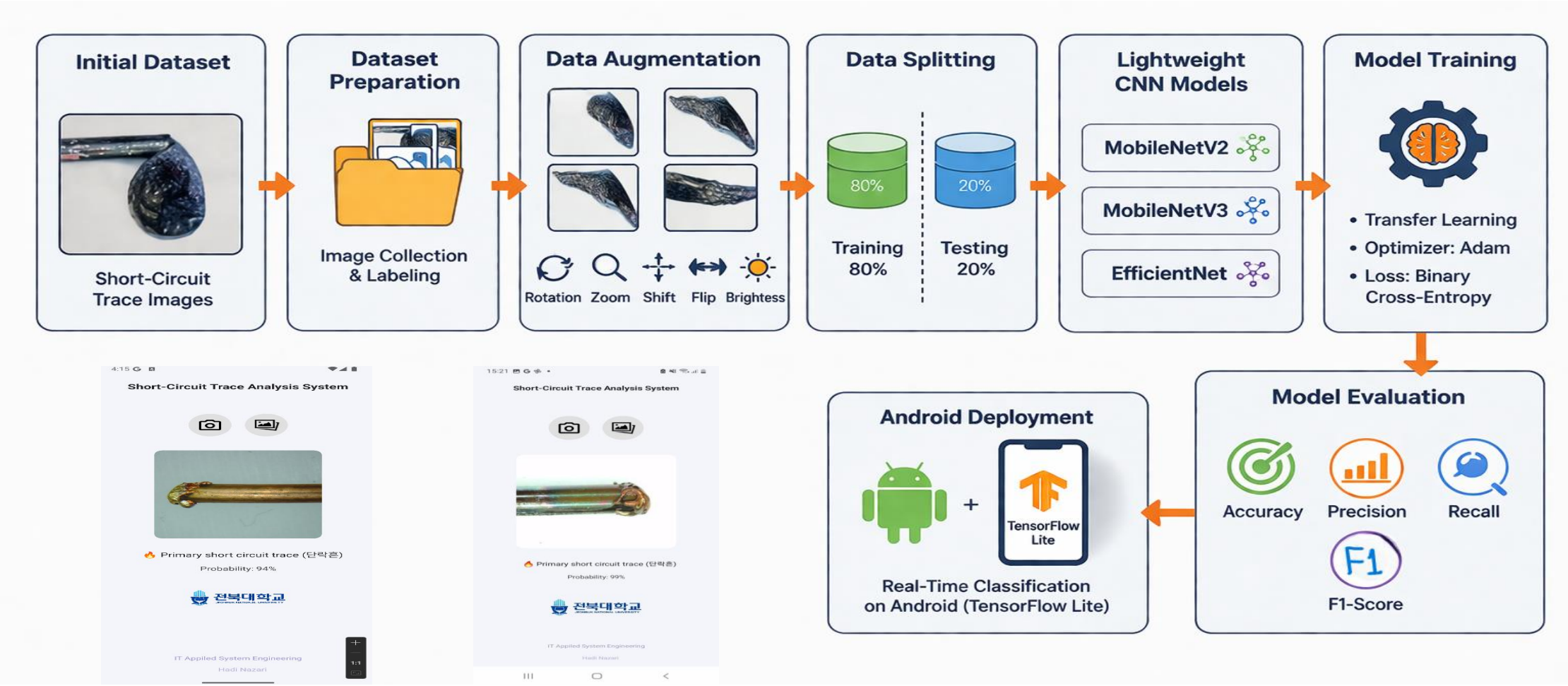


• DEVELOP ANDROID-BASED CLASSIFICATION SYSTEM

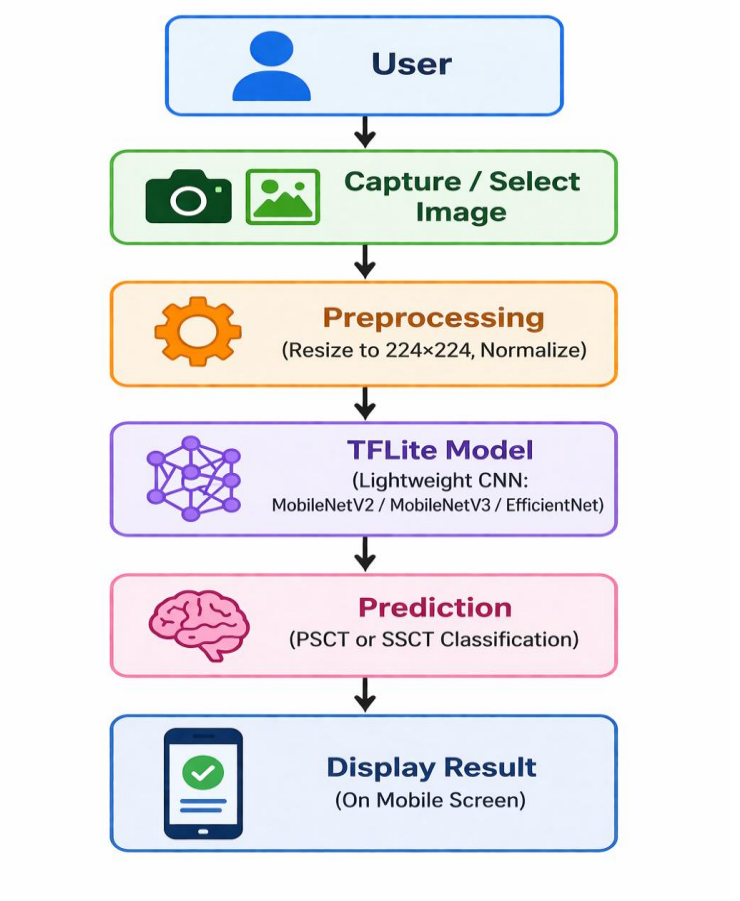
• USE LIGHTWEIGHT CNN MODELS

• ENABLE REAL-TIME MOBILE INFERENCE

PROPOSED FRAMEWORK FOR ELECTRIC FIRE SHORT-CIRCUIT TRACE CLASSIFICATION



ANDROID SYSTEM OVERVIEW



DATASET

Total images: 752

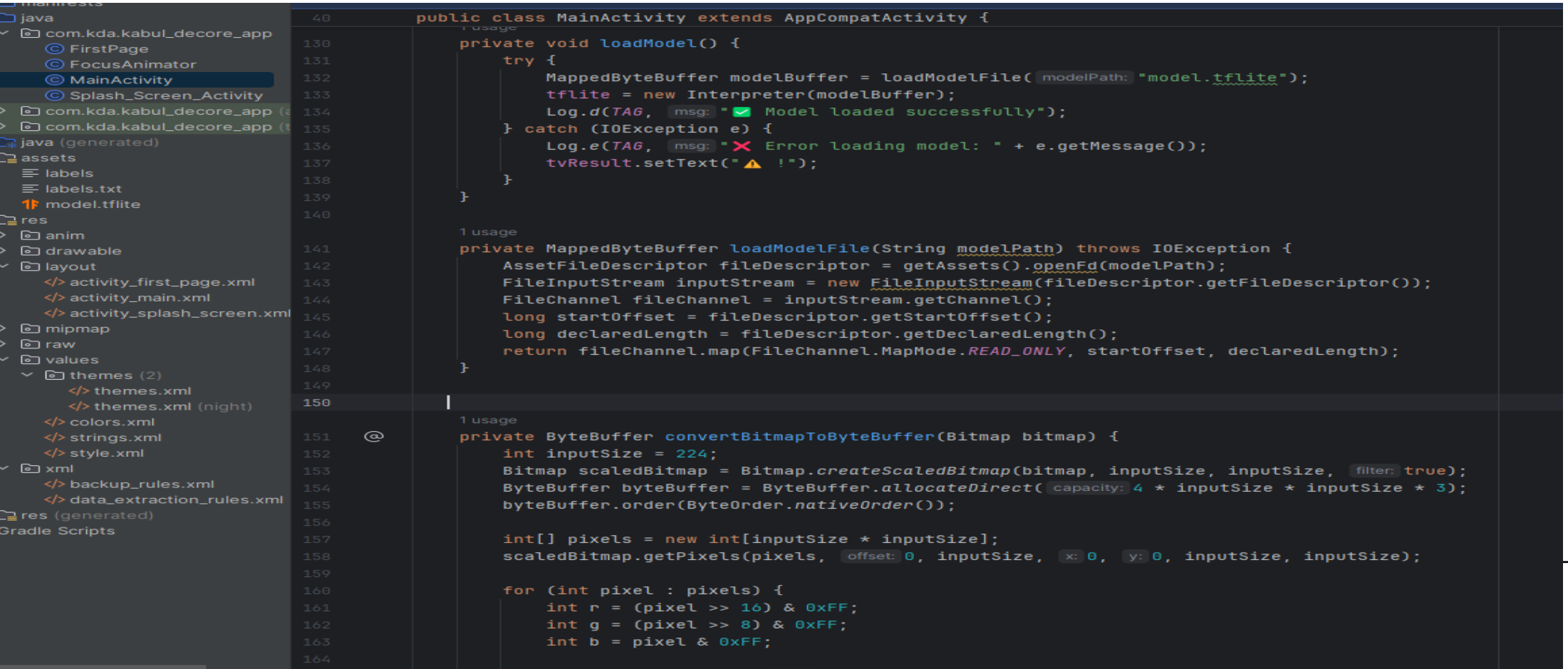
• Classes: primary short-circuit traces (PSCT), secondary short-circuit traces (SSCT)

• Split: 80% training, 20% testing

• Augmentation: rotation, zoom, flip, brightness

PREPROCESSING

- Input image resized to: 224 × 224 pixels
- Converted into ByteBuffer format
- Pixel normalization applied for model compatibility



TensorFlow

MODEL CONVERSION (TENSORFLOW LITE)

• After training, the best-performing model was converted to TensorFlow Lite (.tflite) format.

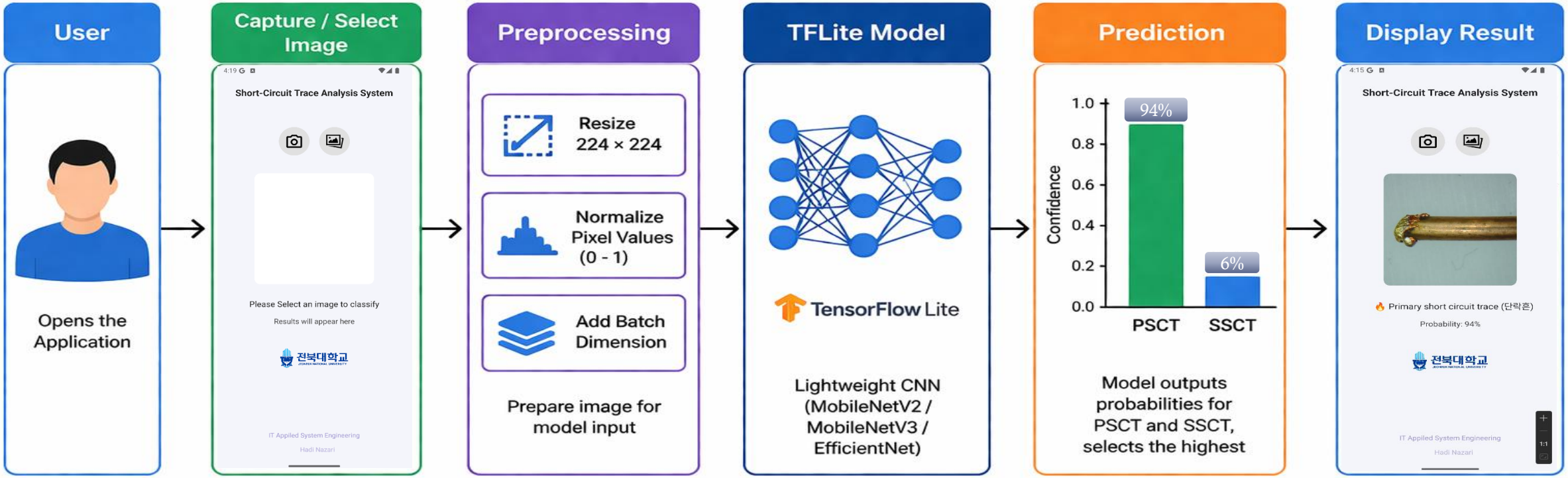
• This step enables:

- •Reduced model size
- •Faster inference
- •Compatibility with mobile devices

ANDROID IMPLEMENTATION

- The mobile application was developed using:
  - **Java**
  - **Android Studio**

SYSTEM OVERVIEW

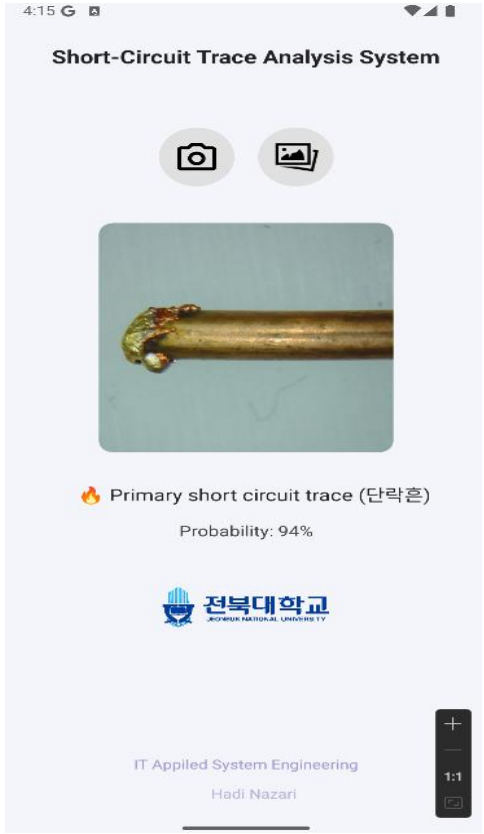


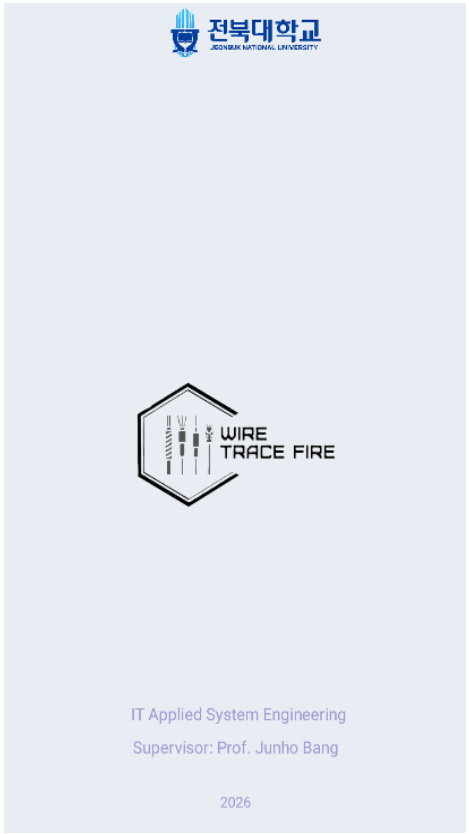
RESULTS

EfficientNet: 96.05% accuracy

- MobileNetV3: stable performance
- MobileNetV2: efficient and fast

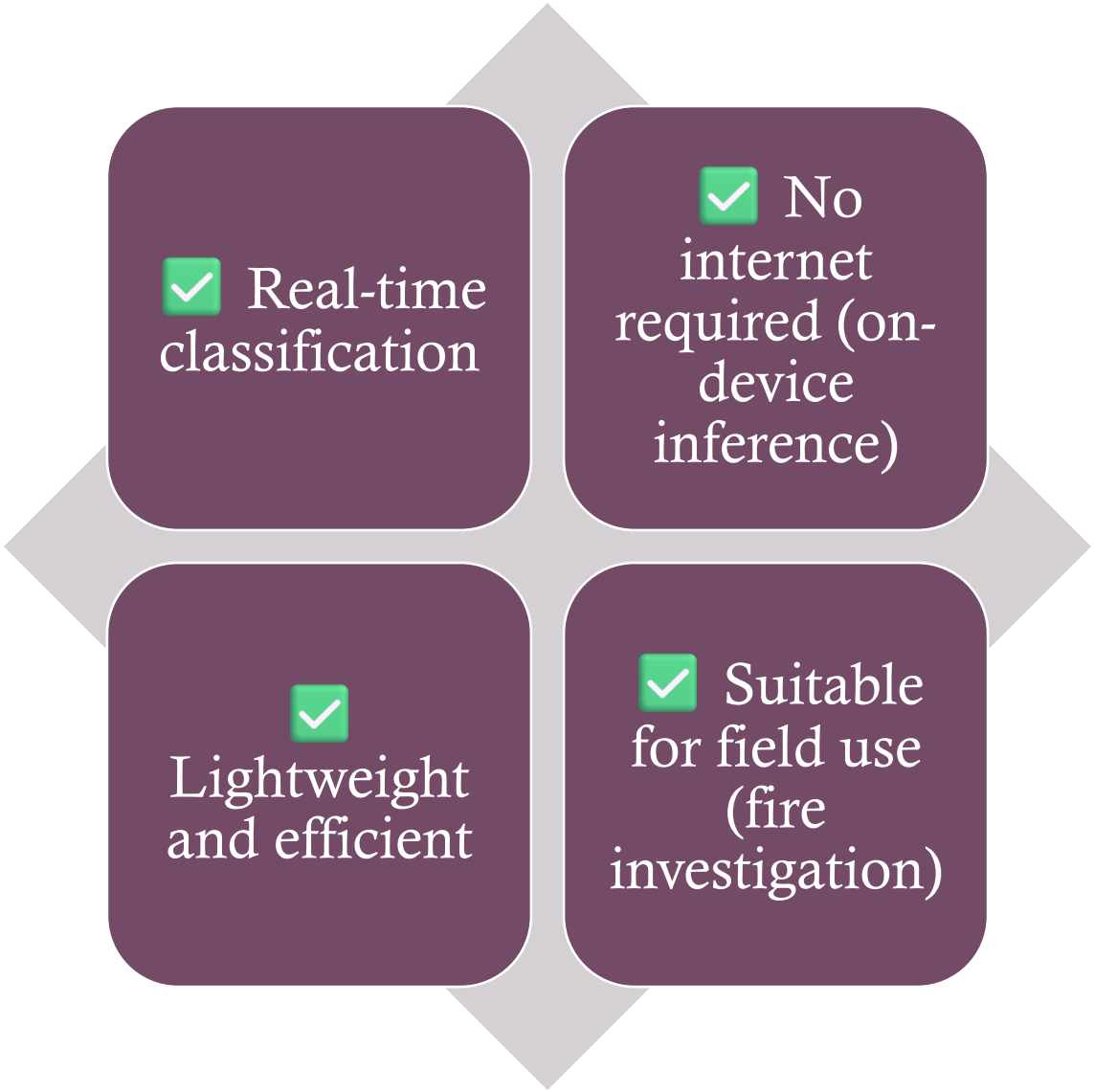
APPLICATION INTERFACE





- Image input (camera/gallery)
- Prediction result
- Probability output
- User-friendly UI

ADVANTAGES OF THE MOBILE IMPLEMENTATION



SUMMARY

- Model trained in Python (Anaconda, TensorFlow/Keras)
- Converted to TensorFlow Lite format
- Integrated into Android Studio (Java)
- Supports camera & gallery input
- Performs real-time on-device classification
- Displays predicted class and confidence